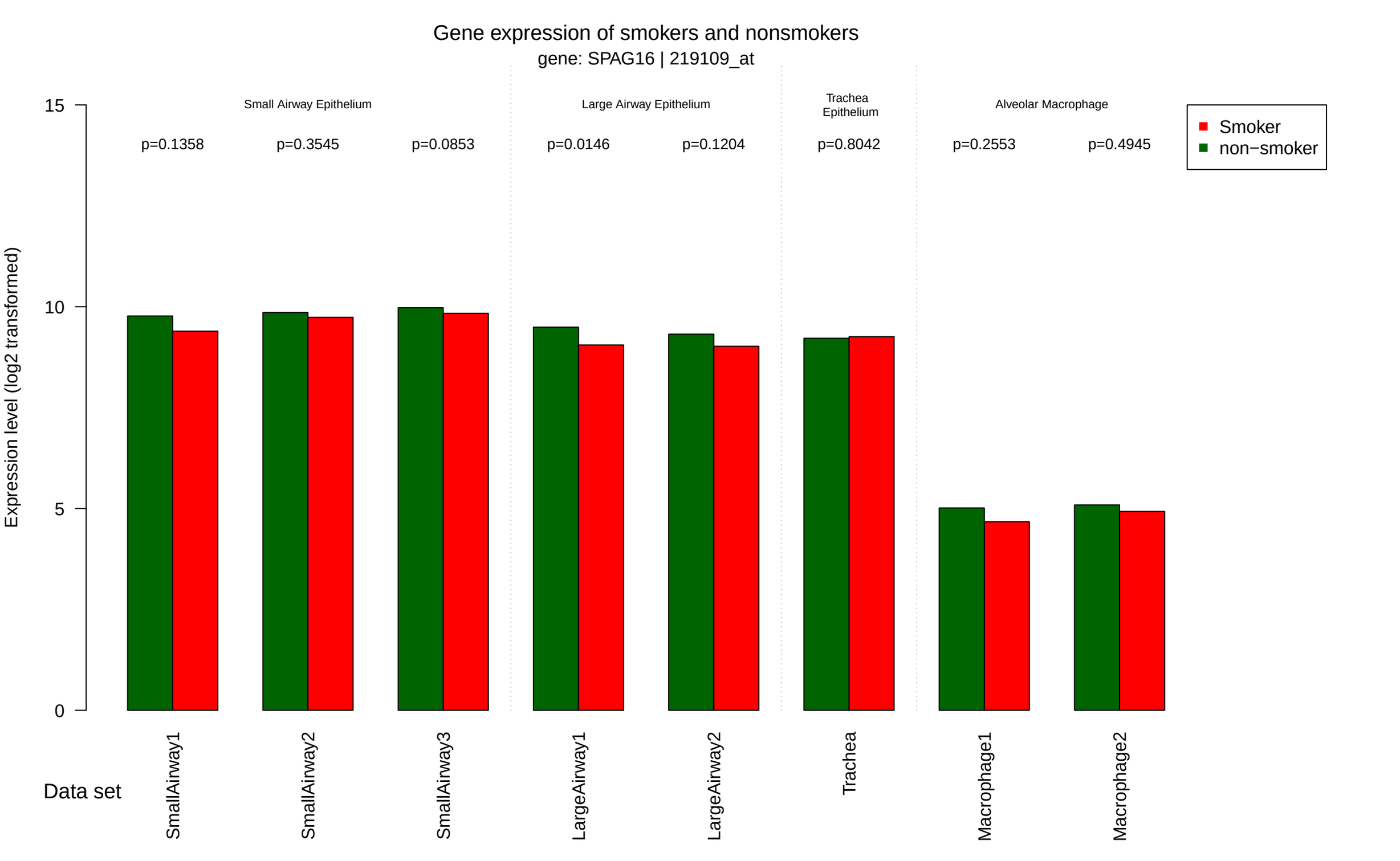
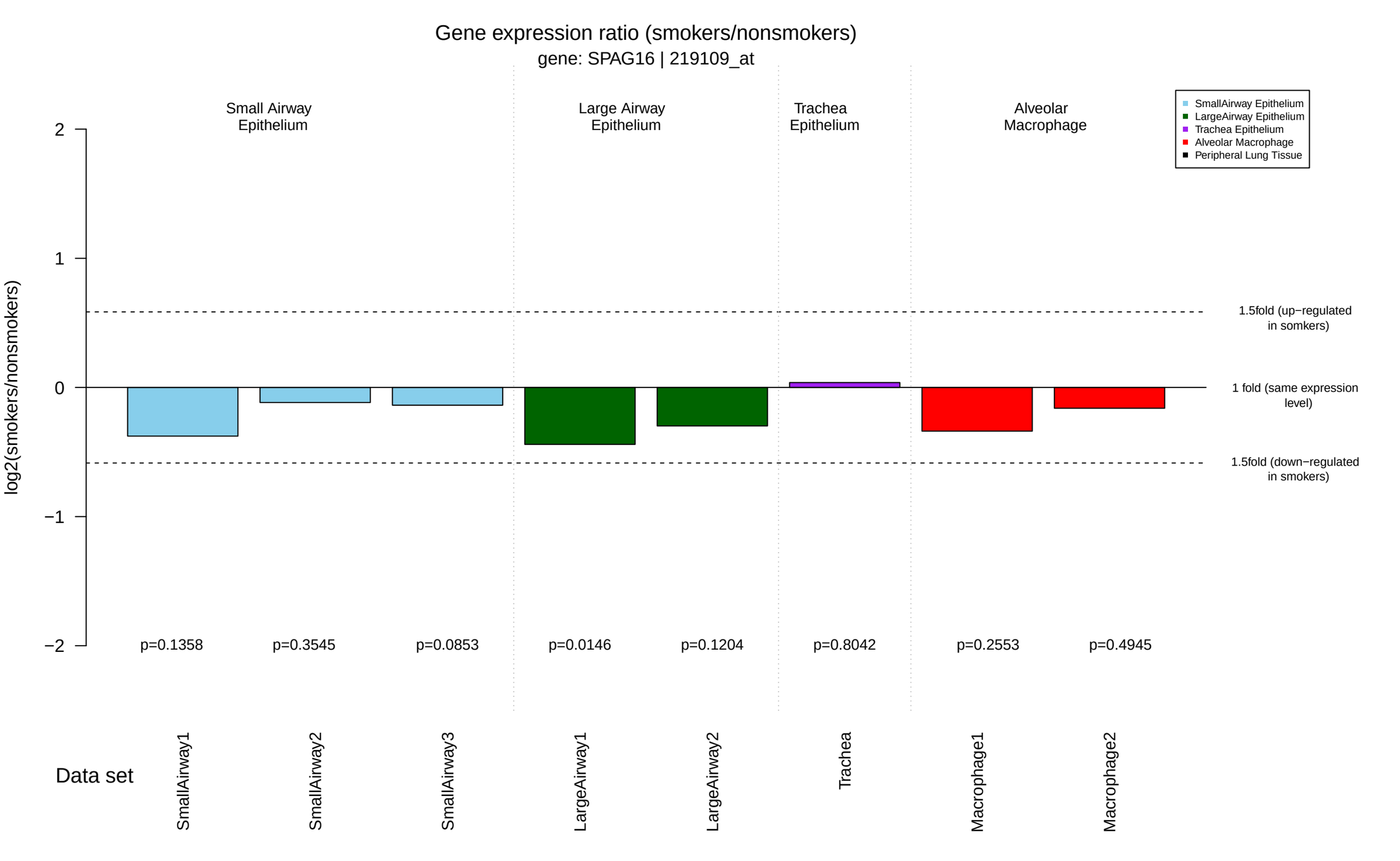
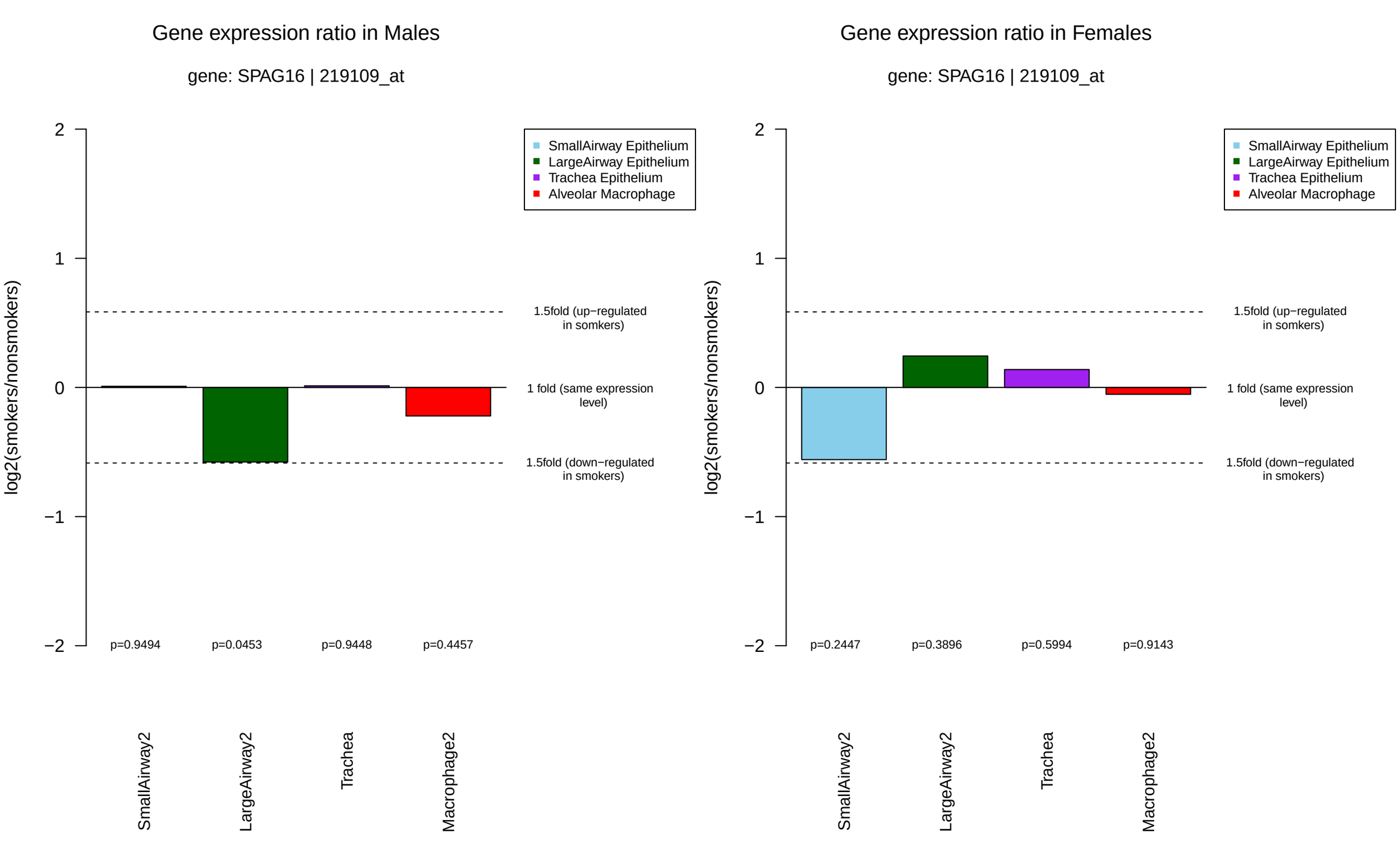
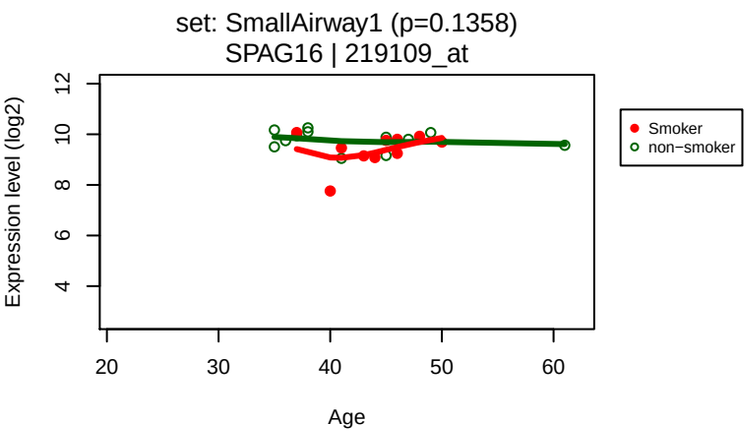
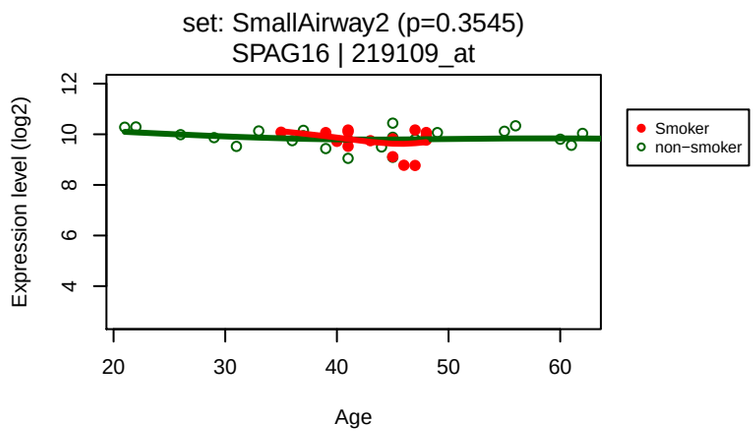
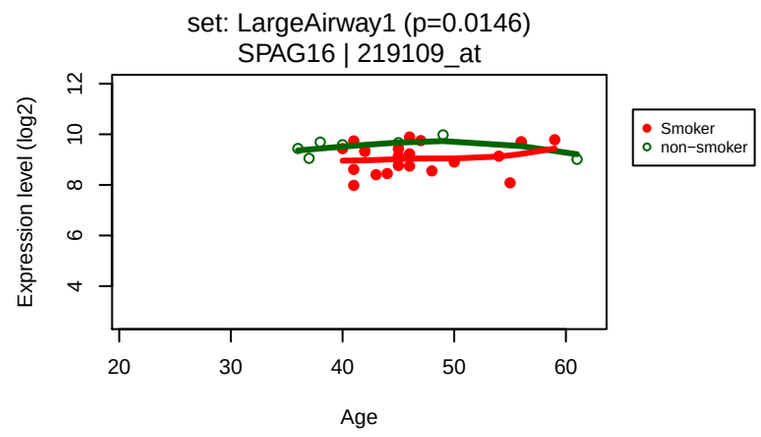

Gene expression of smokers and nonsmokers
gene: SPAG16 | 219109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1358
p=0.3545
p=0.0853
p=0.0146
p=0.1204
p=0.8042
p=0.2553
p=0.4945
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

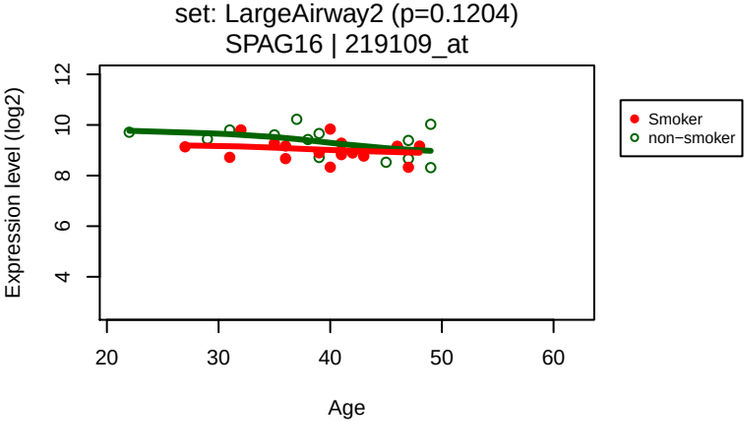

Gene expression ratio (smokers/nonsmokers)
gene: SPAG16 | 219109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1358
p=0.3545
p=0.0853
p=0.0146
p=0.1204
p=0.8042
p=0.2553
p=0.4945
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

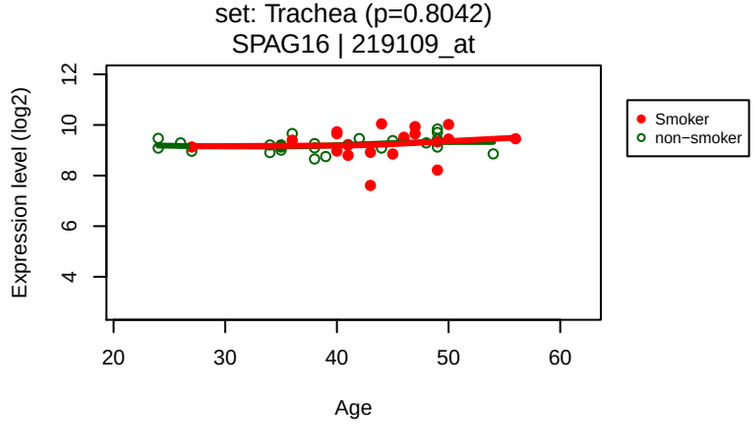

Gene expression ratio in Males
gene: SPAG16 | 219109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9494
p=0.0453
p=0.9448
p=0.4457
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAG16 | 219109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2447
p=0.3896
p=0.5994
p=0.9143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

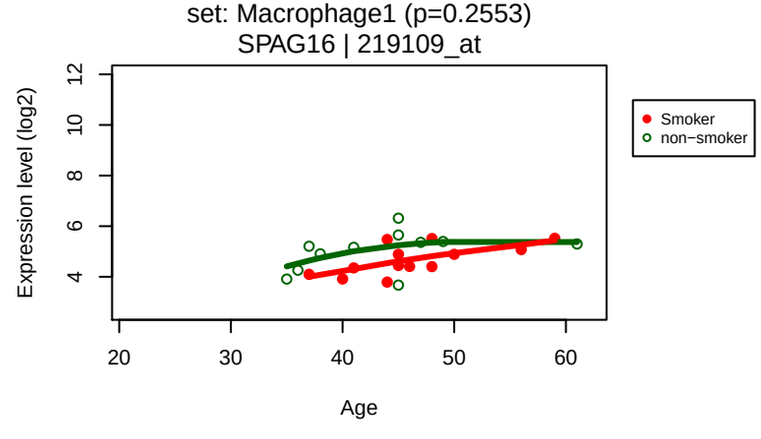

set: SmallAirway1 (p=0.1358)
SPAG16 | 219109_at
Expression level (log2)
Age
Smoker
non−smoker

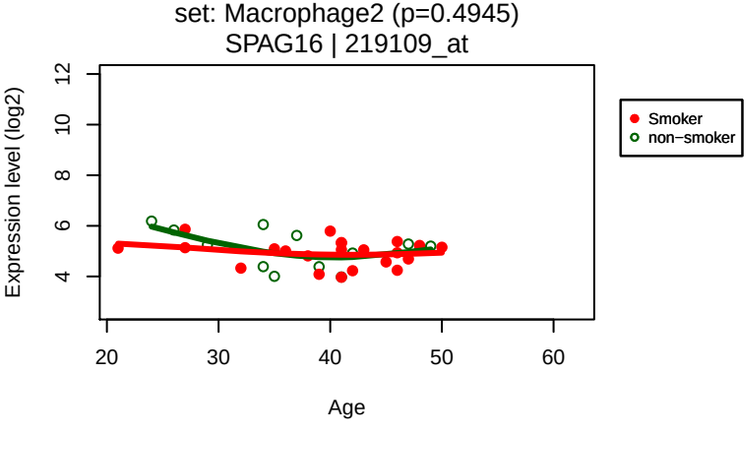

set: SmallAirway2 (p=0.3545)
SPAG16 | 219109_at
Expression level (log2)
Age
Smoker
non−smoker

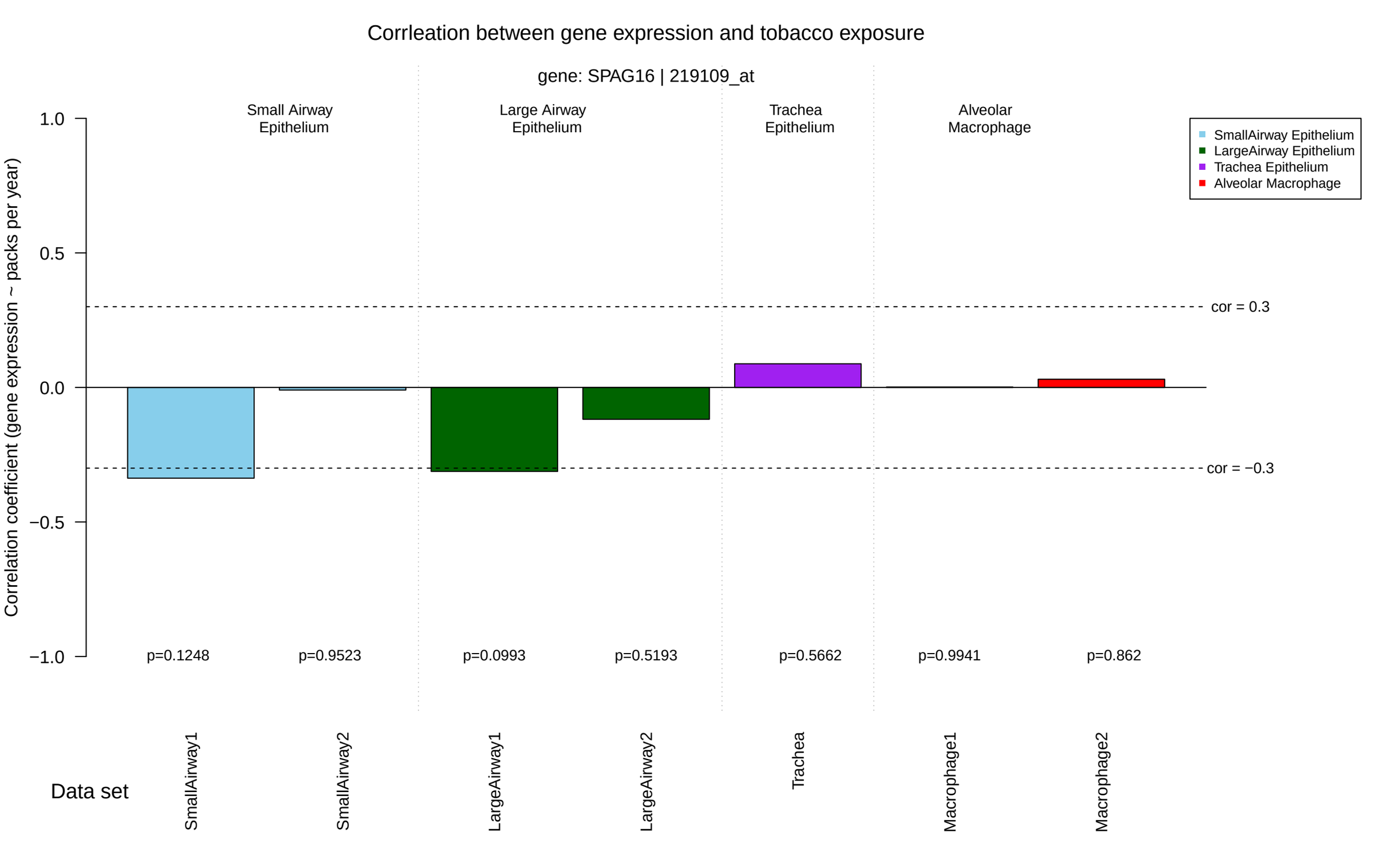

set: LargeAirway1 (p=0.0146)
SPAG16 | 219109_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1204)
SPAG16 | 219109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8042)
SPAG16 | 219109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2553)
SPAG16 | 219109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4945)
SPAG16 | 219109_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAG16 | 219109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1248
p=0.9523
p=0.0993
p=0.5193
p=0.5662
p=0.9941
p=0.862
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2